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1  MySQL database named school and a table named students with the following structure:
2  id (INT)
3  name (VARCHAR)
4  age (INT)
5  grade (VARCHAR)
6
7  import java.sql.*;
8  public class DatabaseMetaDataExample {
9      public static void main(String[] args) {
10         Connection connection = DriverManager.getConnection(
11             "jdbc:mysql://localhost:3306/school";, "root", "");
12         DatabaseMetaData metaData = connection.getMetaData();
13
14         // Print general database information
15         System.out.println("Database Product Name: " + metaData.getDatabaseProductName());
16         System.out.println("Database Product Version: " + metaData.getDatabaseProductVersion());
17         System.out.println("Driver Name: " + metaData.getDriverName());
18         System.out.println("Driver Version: " + metaData.getDriverVersion());
19         System.out.println("URL: " + metaData.getURL());
20         System.out.println("User Name: " + metaData.getUserName());
21
22         // Get tables metadata
23         ResultSet tables = metaData.getTables(null, null, "%", new String[] {"TABLE"});
24         System.out.println("\nTables in the database:");
25         while (tables.next()) {
26             String tableName = tables.getString("TABLE_NAME");
27             System.out.println("Table: " + tableName);
28         }
29
30         // Get columns metadata for a specific table, e.g., 'students'
31         ResultSet columns = metaData.getColumns(null, null, "students", null);
32         System.out.println("\nColumns in the 'students' table:");
33         while (columns.next()) {
34             String columnName = columns.getString("COLUMN_NAME");
35             String columnType = columns.getString("TYPE_NAME");
36             int columnSize = columns.getInt("COLUMN_SIZE");
37             System.out.println("Column: " + columnName + ", Type: " + columnType + ", Size: "
38                 + columnSize);
39         }
40     }
41 }
42
43 Database Product Name: MySQL
44 Database Product Version: 8.0.23
45 Driver Name: MySQL Connector Java
46 Driver Version: 8.0.23
47 URL: jdbc:mysql://localhost:3306/school
48 User Name: root
49 Tables in the database:
50 Table: students
51
52 Columns in the 'students' table:
53 Column: id, Type: INT, Size: 11
54 Column: name, Type: VARCHAR, Size: 255
55 Column: age, Type: INT, Size: 11
56 Column: grade, Type: VARCHAR, Size: 255
57 =====

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53 import java.sql.*;
54
55
56 public class ResultSetMetaDataExample {
57     public static void main(String[] args) {
58         String url = "jdbc:mysql://localhost:3306/school";
59         String username = "root";
60         String password = "password";
61         Connection connection = DriverManager.getConnection(url, username, password);
62         Statement statement = connection.createStatement();
63         ResultSet resultSet = statement.executeQuery("SELECT * FROM students");
64         ResultSetMetaData metaData = resultSet.getMetaData();
65         int columnCount = metaData.getColumnCount();
66
67         // Print column information
68         for (int i = 1; i <= columnCount; i++) {
69             String columnName = metaData.getColumnName(i);
70             String columnType = metaData.getColumnTypeName(i);
71             int columnDisplaySize = metaData.getColumnDisplaySize(i);
72
73             System.out.println("Column " + i + ":");
74             System.out.println("Name: " + columnName);
75             System.out.println("Type: " + columnType);
76             System.out.println("Display Size: " + columnDisplaySize);
77             System.out.println();
78         }
79     }
80 }
81
82 Column 1:
83 Name: id
84 Type: INT
85 Display Size: 11
86
87 Column 2:
88 Name: name
89 Type: VARCHAR
90 Display Size: 255
91
92 Column 3:
93 Name: age
94 Type: INT
95 Display Size: 11
96
97 Column 4:
98 Name: grade
99 Type: VARCHAR
100 Display Size: 255

```